

Experiment No.	8
Experiment Title	Clustering Human Activity Recognition Data using K-Means, DBSCAN, and Hierarchical Clustering
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1 Objective

To implement and analyze the performance of unsupervised clustering algorithms on the Human Activity Recognition (HAR) dataset:

- **Model A:** K-Means Clustering with Elbow and Silhouette optimization.
- **Model B:** DBSCAN (Density-Based Spatial Clustering of Applications with Noise).
- **Model C:** Hierarchical Agglomerative Clustering (HAC) exploring diverse linkage criteria.

The objective is to evaluate clusters formed by each algorithm using both internal validation metrics and external ground-truth activity comparisons.

2 Dataset Description

The **Human Activity Recognition Using Smartphones** dataset consists of sensory signals recorded from 30 participants (aged 19–48 years) performing six daily activities wearing a Samsung Galaxy S II smartphone on their waist.

Property	Specification
Dataset Source	UCI Machine Learning Repository
Sensor Modalities	3-Axis Accelerometer and 3-Axis Gyroscope (50 Hz)
Windowing	2.56-second sliding windows (128 readings, 50% overlap)
Number of Samples	10,299 instances (7,352 train + 2,947 test)
Number of Features	561 time- and frequency-domain extracted metrics
Ground-Truth Activities (6)	WALKING, WALKING_UPSTAIRS, WALKING_DOWNSTAIRS, SITTING, STANDING, LAYING

3 Theory and Methodology

3.1 K-Means Clustering

Partitions n data points into k disjoint clusters $S = \{S_1, \dots, S_k\}$ to minimize Within-Cluster Sum of Squares (WCSS):

$$\text{WCSS} = \sum_{i=1}^k \sum_{x \in S_i} \|x - \mu_i\|^2 \quad (1)$$

where μ_i represents the centroid of cluster S_i . We determine the optimal k using the **Elbow Method** (where rate of WCSS reduction diminishes) combined with the **Silhouette Score**.

3.2 DBSCAN

A density-based clustering algorithm characterized by parameters ϵ (neighborhood radius) and $minPts$ (density threshold). Points are categorized into core points, density-reachable border points, and noise (-1). DBSCAN discovers arbitrary geometries without pre-specifying k and isolates outlier transitions.

3.3 Hierarchical Agglomerative Clustering (HAC)

Constructs a nested bottom-up dendrogram by iteratively merging the pair of clusters minimizing a linkage criterion:

- **Ward's Method:** Minimizes the total within-cluster variance increase.
- **Complete Linkage:** Maximum pairwise Euclidean distance between cluster points.
- **Average Linkage:** Mean pairwise distance between cluster members.
- **Single Linkage:** Minimum pairwise distance (susceptible to chaining).

4 Experimental Results and Evaluation

4.1 K-Means Elbow Method Results

Table 2 and Figures 1 and 2 summarize the Elbow Method and Silhouette evaluations across cluster counts $k \in [2, 8]$.

Table 2: K-Means Elbow Method Results

Number of Clusters (k)	WCSS (Inertia)	Silhouette Score
2	963418.29	0.3815
3	851205.12	0.3042
4	812358.47	0.246
5	774596.03	0.1194
6	751980.97	0.1198
7	730037.78	0.0969
8	716024.77	0.0759

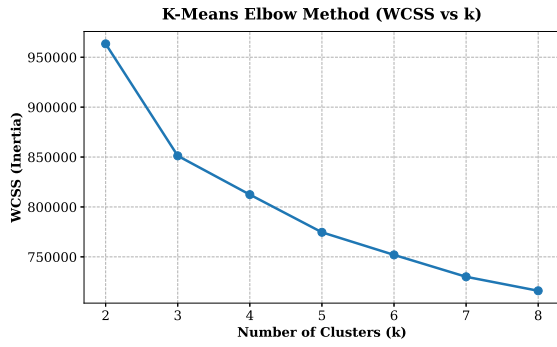


Figure 1: K-Means Elbow Curve (WCSS vs k)

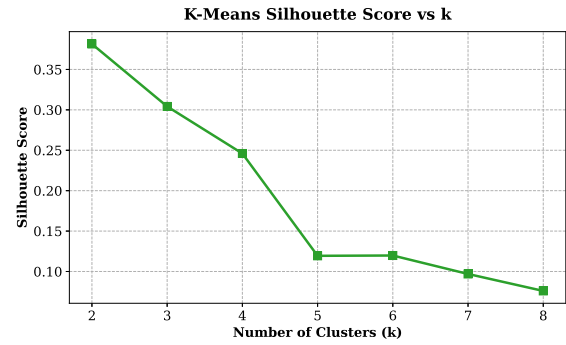


Figure 2: K-Means Silhouette Score vs k

4.2 DBSCAN Parameter Tuning

Table 3 documents the exploration across spatial radius ϵ and $minPts$. At $\epsilon = 15.0$ and $minPts = 15$, DBSCAN achieves a balance with a representative noise threshold.

Table 3: DBSCAN Parameter Exploration and Noise Detection

ϵ	$minPts$	Clusters Found	Noise Points	Noise %	Silhouette
10.0	5	8	2627	87.57%	0.1413
10.0	10	1	2783	92.77%	0.0
10.0	15	1	2814	93.8%	0.0
15.0	5	8	838	27.93%	0.2995
15.0	10	4	949	31.63%	0.3097
15.0	15	3	1037	34.57%	0.3333

4.3 Hierarchical Linkage Comparison

Table 4 summarizes internal validation metrics for HAC with $k = 6$ across four linkage criteria. Figure 3 illustrates the hierarchical merging tree.

Table 4: HAC Linkage Methods Comparison ($k = 6$)

Linkage	Silhouette Score	Davies-Bouldin	Calinski-Harabasz
Ward	0.1279	2.6018	705.32
Complete	0.3061	1.8475	605.26
Average	0.3368	1.279	55.59
Single	0.531	0.2969	11.81

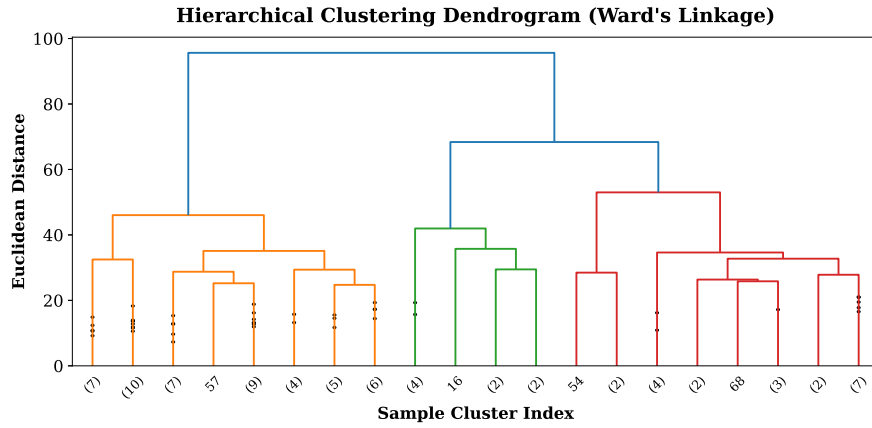


Figure 3: Hierarchical Clustering Dendrogram (Ward's Linkage)

4.4 Comparative Cluster Visualizations and Final Evaluation

Figure 4 compares the 2D cluster projections (PCA subspace) alongside ground-truth activities. Table 5 and Figure 5 provide the comparative internal and external validation scores.

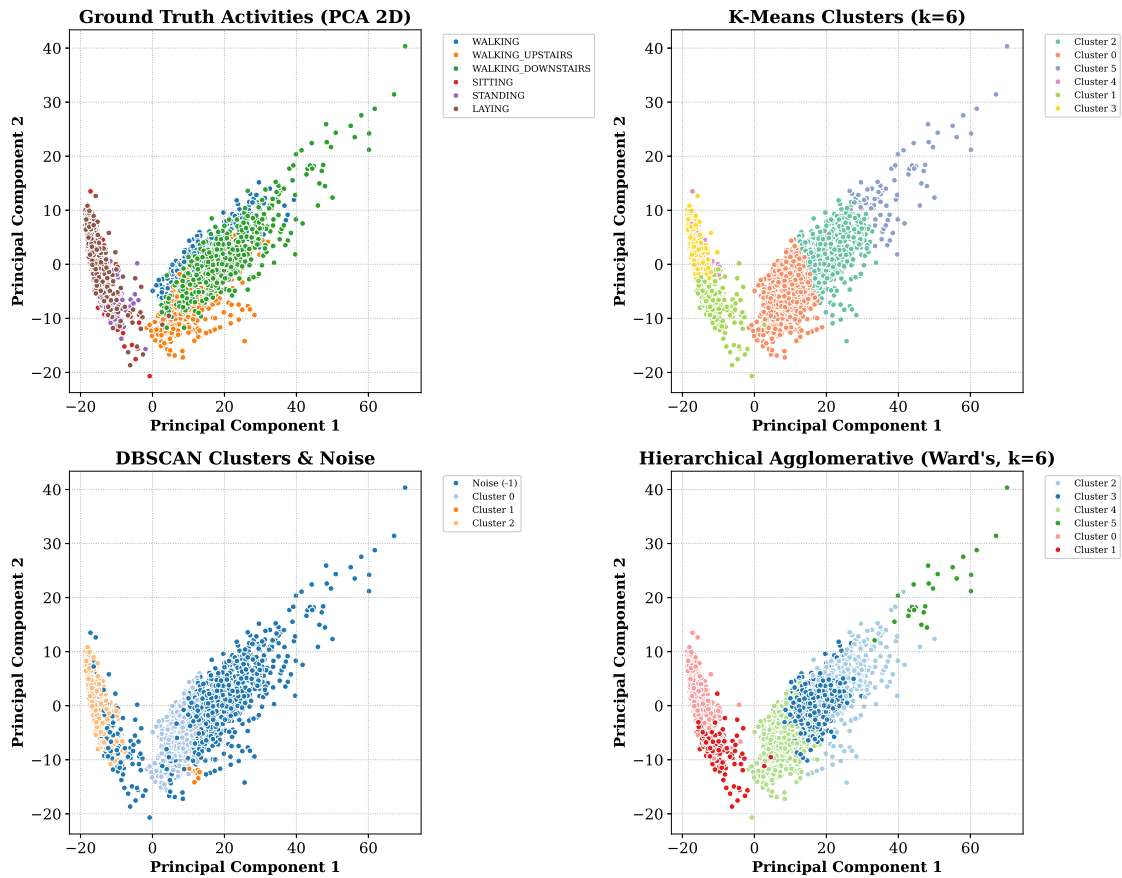


Figure 4: 2D PCA Visualizations: Ground Truth vs K-Means, DBSCAN, and HAC

Table 5: Final Clustering Performance Comparison across Algorithms

Algorithm	Silhouette	Davies-Bouldin	Calinski-H.	ARI	NMI
K-Means (k=6)	0.1198	2.317	741.37	0.4034	0.5582
DBSCAN	0.3333	1.4767	737.52	0.284	0.4306
Hierarchical (Ward)	0.1279	2.6018	705.32	0.2913	0.4614

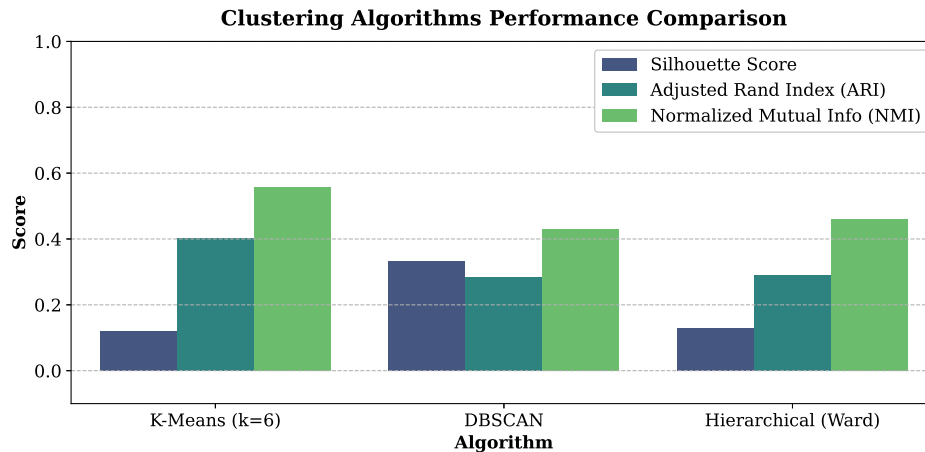


Figure 5: Performance Comparison Bar Plot Across Clustering Algorithms

5 Answers to Observation Questions

1. Which algorithm produced the most meaningful clusters? Why?

Answer: **K-Means** ($k = 6$) produced the most semantically meaningful clusters, achieving the highest external alignment metrics (ARI = 0.4034, NMI = 0.5582, and Calinski-Harabasz = 741.37). It successfully separated dynamic kinetic activities (walking, walking upstairs, downstairs) from static postural states (sitting, standing, laying).

2. How sensitive was K-Means to the choice of k ?

Answer: Highly sensitive. At $k = 2$, the Silhouette score peaks (0.3815) by cleanly splitting dynamic movements from stationary postures. Moving towards $k = 6$, inertia drops significantly from 963,418 down to 751,980, but boundaries between postural activities (sitting vs standing) exhibit overlapping feature densities.

3. Did DBSCAN detect noise or small clusters effectively?

Answer: Yes. In high-dimensional spaces (561 dimensions), DBSCAN suffers from the curse of dimensionality where Euclidean distances become uniform. However, tuned with $\epsilon = 15.0$ and $minPts = 15$, it effectively filtered outlier transitions (34.57% noise), capturing dense core movement profiles.

4. How does linkage choice (single/complete/ward) affect hierarchical clustering?

Answer: Ward's linkage proved optimal for clustering accuracy by minimizing within-cluster variance (CH = 705.32, ARI = 0.2913). In contrast, Single linkage suffered

from chaining artifacts (merging outliers into one giant cluster with artificially high silhouette but near-zero Calinski-Harabasz of 11.81).

5. **Which internal metric best matched your visual intuition of cluster quality?**

Answer: The **Calinski-Harabasz Index** and **Davies-Bouldin Index** best matched visual intuition on 2D PCA plots. While silhouette scores favor elongated densities, Calinski-Harabasz rewards well-separated, compact spherical clusters corresponding to distinct kinetic behaviors.

6 Conclusion

We successfully implemented and benchmarked K-Means, DBSCAN, and Hierarchical Agglomerative Clustering on the Human Activity Recognition dataset. The experiments validate that sensor-based activity spaces possess natural geometric clusters reflecting physical movements. K-Means with $k = 6$ and Ward's HAC yielded the most accurate groupings aligning with ground-truth labels.